52. Probing and Lightweight Fine-Tuning of DNA Foundation Models

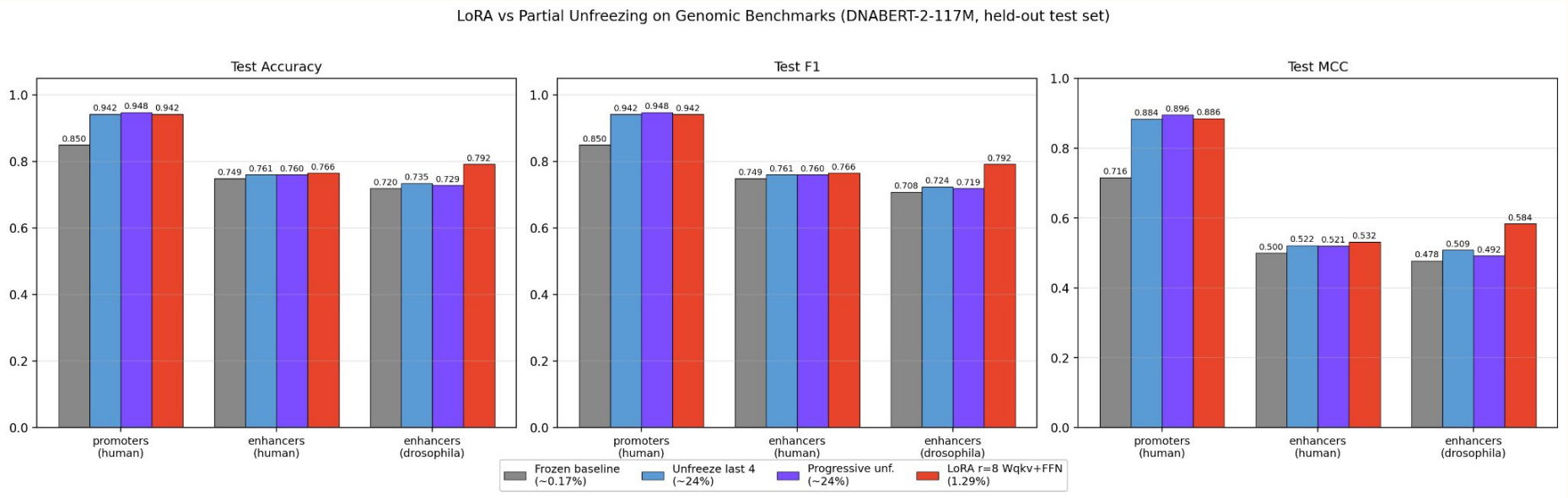
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Experiment Setup

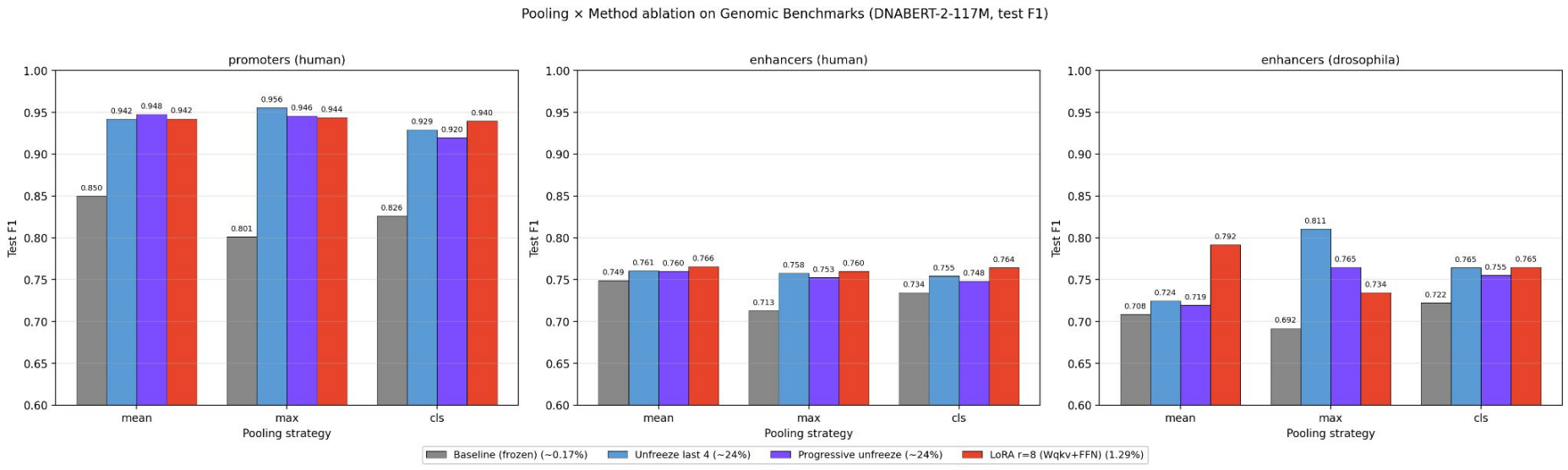
- **Datasets: Genomic Benchmarks:**
  - Human non-TATA promoters — promoter vs. non-promoter sequences
  - Human enhancers (Cohn) — human enhancer region identification
  - Drosophila enhancers (Stark) — cross-species enhancer detection in fruit fly genome
- **Models:**
  - DNABERT-2 — efficient transformer using Byte Pair Encoding
  - Nucleotide Transformer — large BERT-style model pretrained on multi-species genomic data
  - HyenaDNA — convolution-based model without attention pretrained on human reference genome data
- **Experiments:**
  - Frozen probing - model weights fully frozen
  - Partial unfreezing - top transformer layers unfrozen and fine-tuned end-to-end
  - LoRA fine-tuning - low-rank adapters injected into attention layers

DNABERT2



- Frozen Baseline is already strong
- On two human tasks, all fine tuning methods are statistically tied.
- On Drosophila enhancers(cross-species), LoRA get best score.

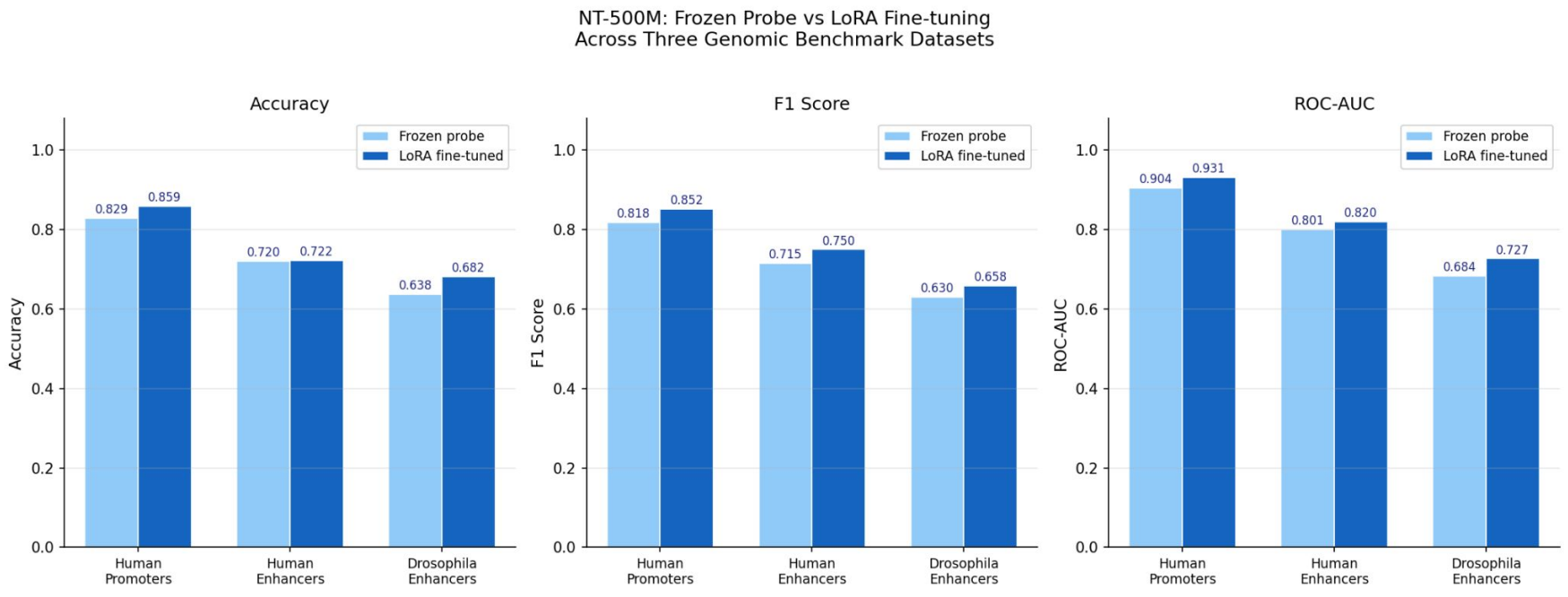
DNABERT2



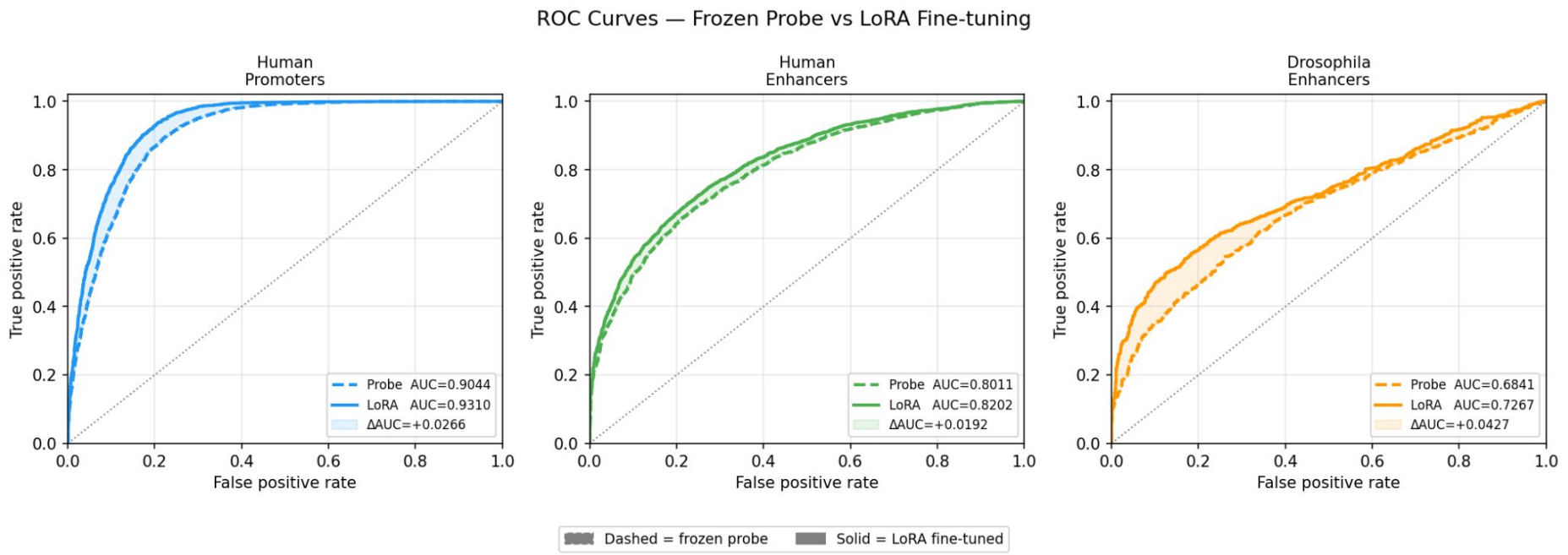
- Drosophila shows the difference between LoRA and Partial unfreezing
- LoRA is benefited from mean pooling strategy
- Partial unfreezing is benefited from max pooling strategy

Nucleotide Transformer

LoRA fine-tuning wins in performance across the board

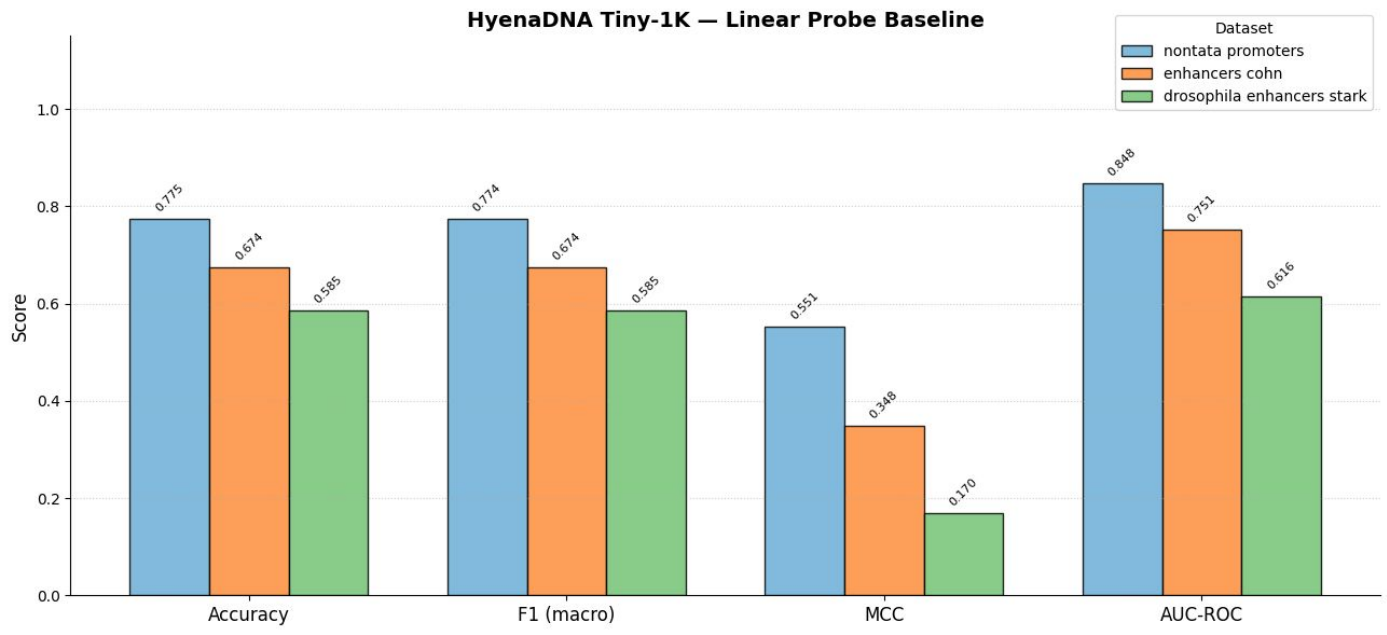


Nucleotide Transformer



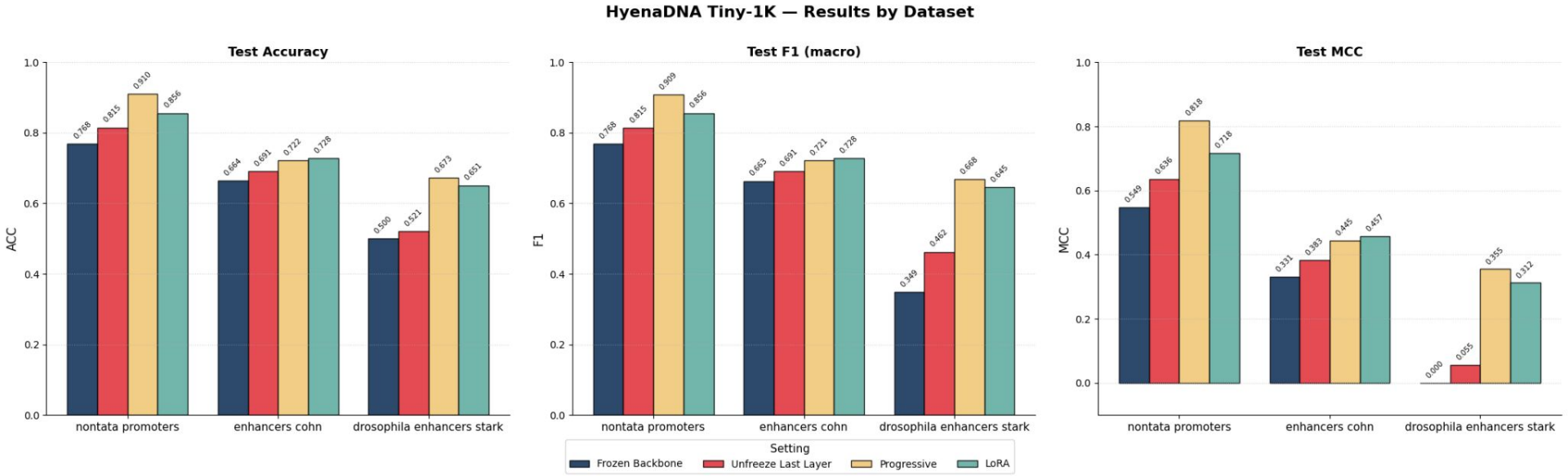
HyenaDNA

- Decoder-only model:
  - Tiny-1k
  - Small 32-k

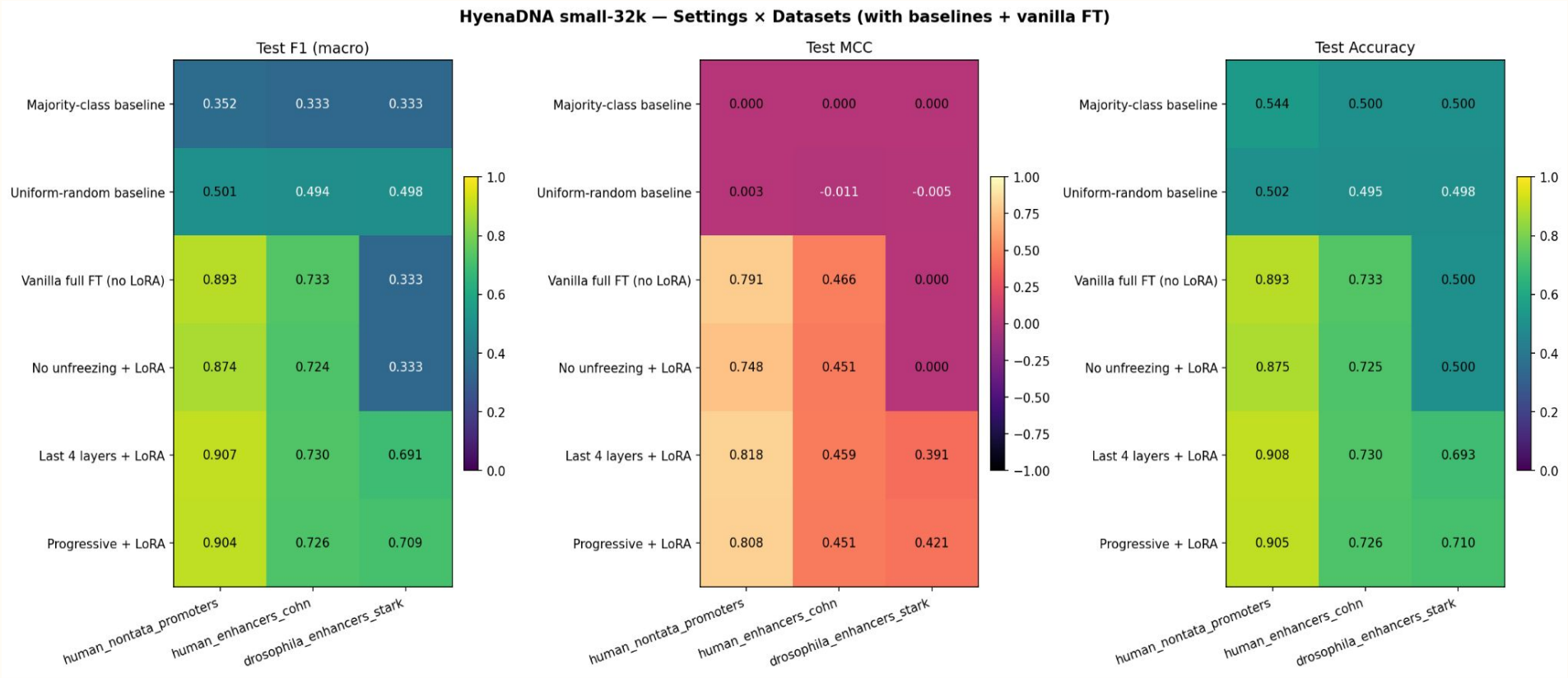


HyenaDNA-1k Results

- Progressive unfreezing achieves highest performance (0.909 F1/ 0.818 MCC)
- LoRA is competitive and most parameter efficient
- Performance decreases with dataset difficulty



HyenaDNA-32k



Conclusion

- Frozen probing is a strong, cheap starting point
- LoRA offers the best performance-efficiency trade-off
- Higher rank generally outperforms lower rank
- Cross-species tasks remain hardest across all models
- No single strategy wins universally — architecture and task both matter
- Future Work:
  - Scale up to larger model variants
  - Test on more diverse genomic tasks
  - Systematic hyperparameter search over LoRA rank, alpha, and target modules